Mecanum Wheeled Robot

Outcome:

To develop a Mecanum wheel robot car that can move in all directions (forward,backward,left,right,forward_left,forward_right,backward_left,backward_right,rotate_left,rotate_right) controlled by a custom joystick-style web interface using ESP32-C6.

Components:

S. No	Component	Quantity
1.	ESP32-C6 Dev Board	1
2.	L298N Motor Driver	2
3.	12v DC Motors(300RPM)	4
4.	Jumper Wires	As needed
5.	USB Type-C Cable	1
6.	Computer with Arduino IDE	1
7.	Robot Chassis	1
8.	12v Battery	1
9.	Mecanum Wheel	4

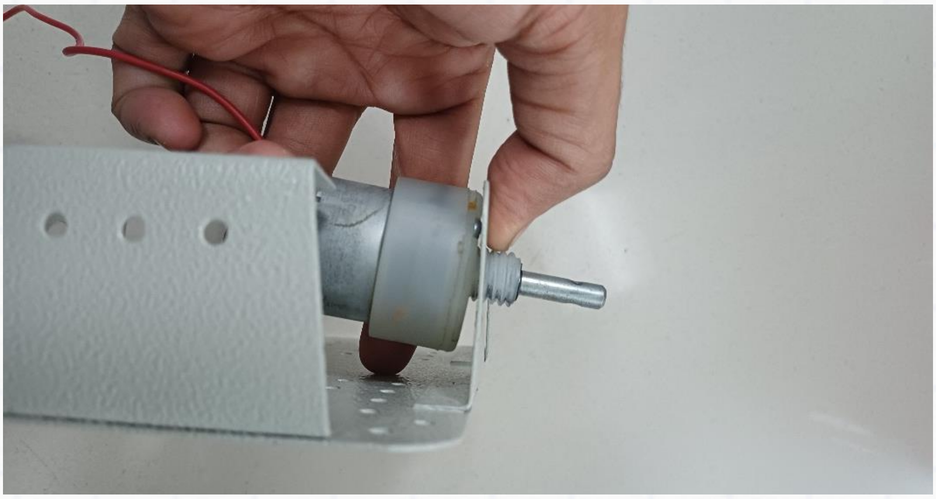
Concept:

The ESP32-C6 is a Wi-Fi-capable microcontroller. In this project, a simple joystick is implemented on a web page hosted by the ESP32-C6. Users interact with the joystick via a browser, which sends directional commands (forward, backward, left,right,forward_left,forward_right,backward_left,backward_right,rotate_left,rotate_right, stop) to the ESP32. The microcontroller then controls two DC motors via an L298N driver based on the received commands, enabling different movements. JavaScript on the client side captures the user's input on a virtual joystick (canvas), detects direction using vector calculations, and sends HTTP GET requests (e.g., /move?dir=forward) to the ESP32 server. The ESP32 interprets these commands and sets motor pins accordingly.

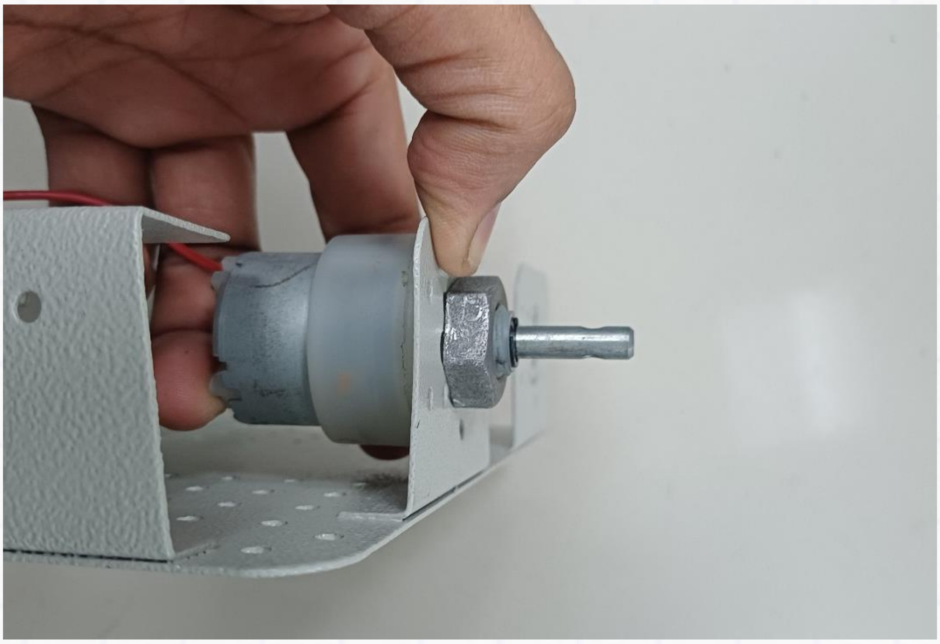
Connections:

1. Hardware Setup:

Follow the steps below to assemble the Motors & Wheels to the Robot Chassis.



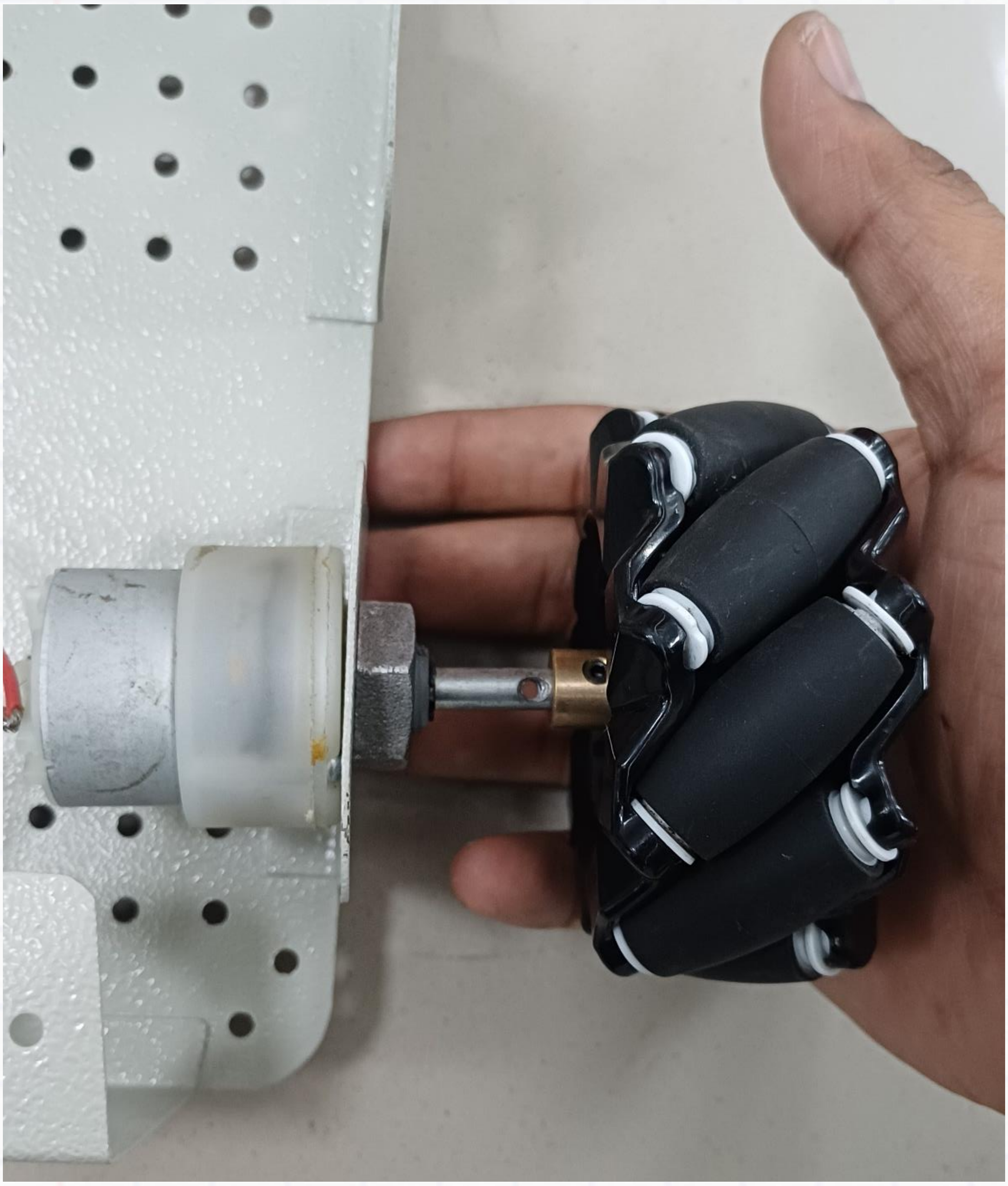
Step 1 – Insert Motor to their respective slots



Step 2 – Use Hexagonal Nut to mount motor & tighten

Step 3 – Insert Brass adapter to Wheel Hub as shown

Step 4 – Insert the provided bolts through the wheel & tighten brass nut

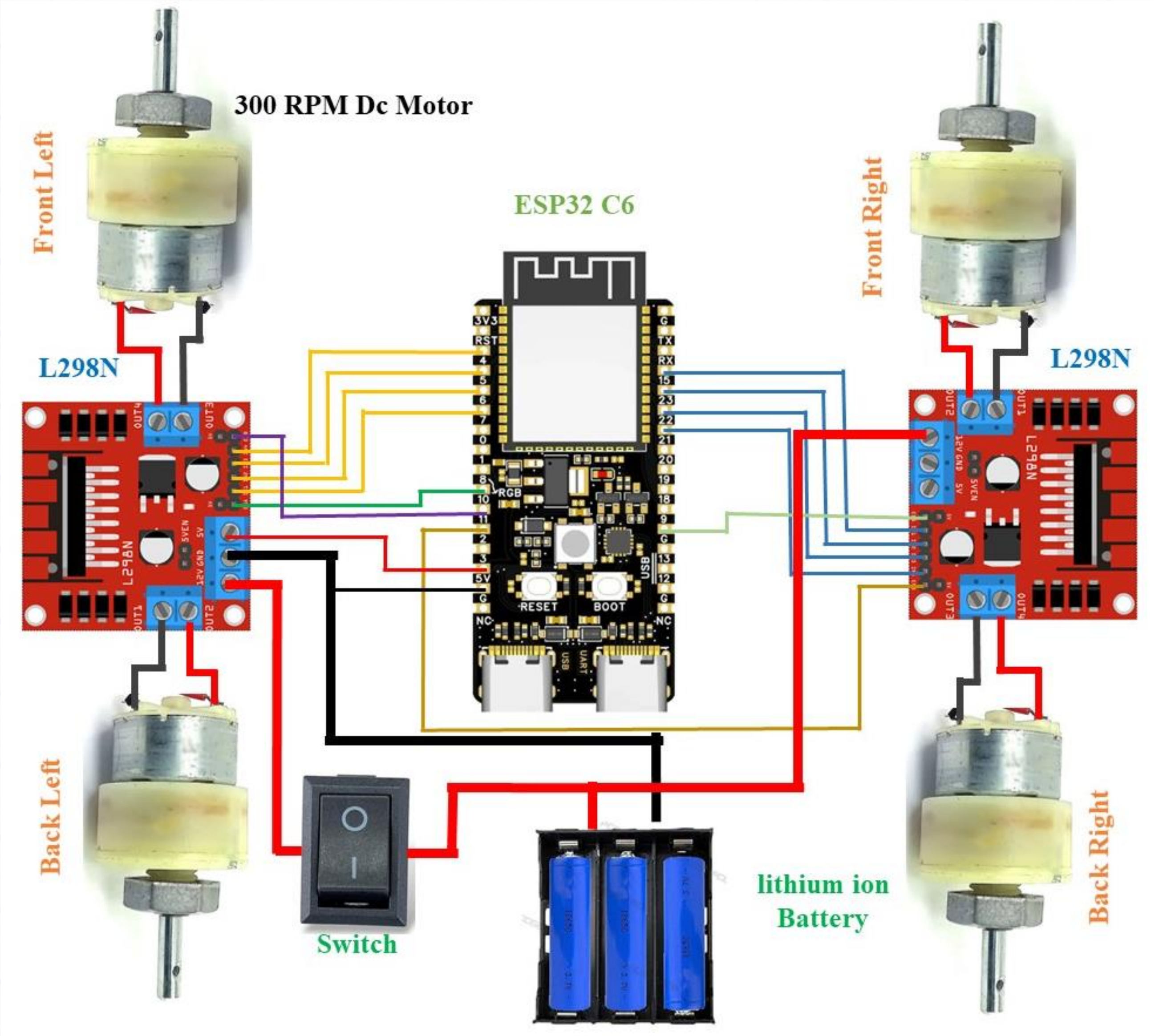


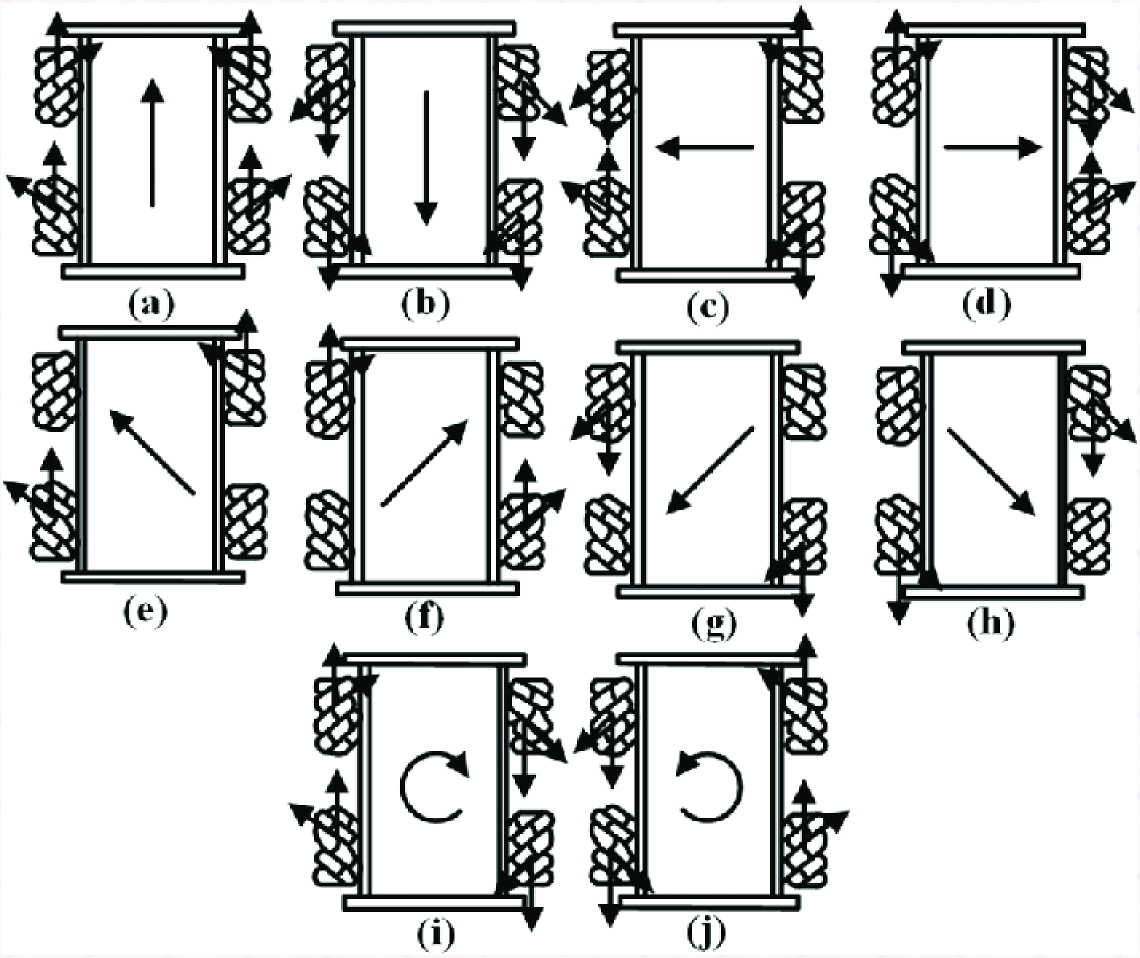
Step 5 – Assemble the Wheel with the motor shaft and tighten the grub screws on the brass adapter

Replicate the same to other three motors & wheels and mount them on the chassis.

1. b) Electrical Setup:
    - Connect 4 motors to the motor driver modules.
    - Wire each driver's input (IN1, IN2, EN) to the ESP32-C6 GPIO pins as per the pinout.
    - Connect motor power supply (5V-12V battery) to the motor drivers.
    - Connect ESP32-C6 to PC via USB for programming.
2. Software Setup:
    - Open Arduino IDE.
    - Install ESP32 board via Board Manager (use latest Espressif version).
    - Select "ESP32C6 Dev Module" under Tools > Board.
    - Paste the given code in Arduino IDE
    - Replace **SSID** and **PASSWORD** with your Wi-Fi credentials
    - Upload the code to the ESP32-C6
    - Open Serial Monitor to view assigned IP address
3. Testing:
    - Open Serial Monitor to confirm Wi-Fi connection and note IP address.
    - Open any browser on the same Wi-Fi network.
    - Visit the ESP32's IP (e.g., http://192.168.4.1 or assigned IP).
    - Use the on-screen controls to move the robot in different directions.
    - Use the speed slider to control motor speed (PWM 0-255).

Circuit Diagram:





Mecanum wheels Diagram

Software Code:

```
#include <WiFi.h>
#include <WebServer.h>

const char* ssid = "1234";
const char* password = "123456789m";

WebServer server(80);

// Motor Direction Pins
#define M1_IN1 15
#define M1_IN2 23
#define M2_IN1 6
#define M2_IN2 7
#define M3_IN1 4
#define M3_IN2 5
#define M4_IN1 21
#define M4_IN2 22

// Motor Enable Pins (PWM)
#define M1_EN 8
#define M2_EN 9
#define M3_EN 10
#define M4_EN 11

int speedPWM = 150;

void stopAll() {
  digitalWrite(M1_IN1, LOW); digitalWrite(M1_IN2, LOW); analogWrite(M1_EN, 0);
  digitalWrite(M2_IN1, LOW); digitalWrite(M2_IN2, LOW); analogWrite(M2_EN, 0);
  digitalWrite(M3_IN1, LOW); digitalWrite(M3_IN2, LOW); analogWrite(M3_EN, 0);
  digitalWrite(M4_IN1, LOW); digitalWrite(M4_IN2, LOW); analogWrite(M4_EN, 0);
}
void move(String dir) {
  stopAll(); // Ensure all motors are off before starting

  if (dir == "forward") {
    digitalWrite(M1_IN1, LOW); digitalWrite(M1_IN2, HIGH); analogWrite(M1_EN, 
speedPWM);
    digitalWrite(M2_IN1, LOW); digitalWrite(M2_IN2, HIGH); analogWrite(M2_EN, 
speedPWM);
    digitalWrite(M3_IN1, LOW); digitalWrite(M3_IN2, HIGH); analogWrite(M3_EN, 
speedPWM);
    digitalWrite(M4_IN1, LOW); digitalWrite(M4_IN2, HIGH); analogWrite(M4_EN, 
speedPWM);
  } else if (dir == "backward") {
    digitalWrite(M1_IN1, HIGH); digitalWrite(M1_IN2, LOW); analogWrite(M1_EN, 
speedPWM);
```

```
    digitalWrite(M2_IN1, HIGH); digitalWrite(M2_IN2, LOW); analogWrite(M2_EN, 
speedPWM);
    digitalWrite(M3_IN1, HIGH); digitalWrite(M3_IN2, LOW); analogWrite(M3_EN, 
speedPWM);
    digitalWrite(M4_IN1, HIGH); digitalWrite(M4_IN2, LOW); analogWrite(M4_EN, 
speedPWM);
  } else if (dir == "left") {
    digitalWrite(M1_IN1, LOW); digitalWrite(M1_IN2, HIGH); analogWrite(M1_EN, 
speedPWM);
    digitalWrite(M2_IN1, HIGH); digitalWrite(M2_IN2, LOW); analogWrite(M2_EN, 
speedPWM);
    digitalWrite(M3_IN1, LOW); digitalWrite(M3_IN2, HIGH); analogWrite(M3_EN, 
speedPWM);
    digitalWrite(M4_IN1, HIGH); digitalWrite(M4_IN2, LOW); analogWrite(M4_EN, 
speedPWM);
  } else if (dir == "right") {
    digitalWrite(M1_IN1, HIGH); digitalWrite(M1_IN2, LOW); analogWrite(M1_EN, 
speedPWM);
    digitalWrite(M2_IN1, LOW); digitalWrite(M2_IN2, HIGH); analogWrite(M2_EN, 
speedPWM);
    digitalWrite(M3_IN1, HIGH); digitalWrite(M3_IN2, LOW); analogWrite(M3_EN, 
speedPWM);
    digitalWrite(M4_IN1, LOW); digitalWrite(M4_IN2, HIGH); analogWrite(M4_EN, 
speedPWM);
  } else if (dir == "forward_left") {
    digitalWrite(M1_IN1, LOW); digitalWrite(M1_IN2, HIGH); analogWrite(M1_EN, 
speedPWM);
    digitalWrite(M2_IN1, LOW); digitalWrite(M2_IN2, LOW); analogWrite(M2_EN, 
speedPWM);
    digitalWrite(M3_IN1, LOW); digitalWrite(M3_IN2, HIGH); analogWrite(M3_EN, 
speedPWM);
    digitalWrite(M4_IN1, LOW); digitalWrite(M4_IN2, LOW); analogWrite(M4_EN, 
speedPWM);
  } else if (dir == "forward_right") {
    digitalWrite(M1_IN1, LOW); digitalWrite(M1_IN2, LOW); analogWrite(M1_EN, 
speedPWM);
    digitalWrite(M2_IN1, LOW); digitalWrite(M2_IN2, HIGH); analogWrite(M2_EN, 
speedPWM);
    digitalWrite(M3_IN1, LOW); digitalWrite(M3_IN2, LOW); analogWrite(M3_EN, 
speedPWM);
    digitalWrite(M4_IN1, LOW); digitalWrite(M4_IN2, HIGH); analogWrite(M4_EN, 
speedPWM);
  } else if (dir == "backward_left") {
    digitalWrite(M1_IN1, LOW); digitalWrite(M1_IN2, LOW); analogWrite(M1_EN, 
speedPWM);
    digitalWrite(M2_IN1, HIGH); digitalWrite(M2_IN2, LOW); analogWrite(M2_EN, 
speedPWM);
```

```
    digitalWrite(M3_IN1, LOW); digitalWrite(M3_IN2, LOW); analogWrite(M3_EN,
speedPWM);
    digitalWrite(M4_IN1, HIGH); digitalWrite(M4_IN2, LOW); analogWrite(M4_EN,
speedPWM);
  } else if (dir == "backward_right") {
    digitalWrite(M1_IN1, HIGH); digitalWrite(M1_IN2, LOW); analogWrite(M1_EN,
speedPWM);
    digitalWrite(M2_IN1, LOW); digitalWrite(M2_IN2, LOW); analogWrite(M2_EN,
speedPWM);
    digitalWrite(M3_IN1, HIGH); digitalWrite(M3_IN2, LOW); analogWrite(M3_EN,
speedPWM);
    digitalWrite(M4_IN1, LOW); digitalWrite(M4_IN2, LOW); analogWrite(M4_EN,
speedPWM);
  } else if (dir == "rotate_left") {
    digitalWrite(M1_IN1, LOW); digitalWrite(M1_IN2,HIGH); analogWrite(M1_EN,
speedPWM);//2
    digitalWrite(M2_IN1, LOW); digitalWrite(M2_IN2, HIGH); analogWrite(M2_EN,
speedPWM);//4
    digitalWrite(M3_IN1, HIGH); digitalWrite(M3_IN2, LOW); analogWrite(M3_EN,
speedPWM);//3
    digitalWrite(M4_IN1, HIGH); digitalWrite(M4_IN2, LOW); analogWrite(M4_EN,
speedPWM);//1
  } else if (dir == "rotate_right") {
    digitalWrite(M1_IN1, HIGH); digitalWrite(M1_IN2,LOW); analogWrite(M1_EN,
speedPWM);//2
    digitalWrite(M2_IN1, HIGH); digitalWrite(M2_IN2, LOW); analogWrite(M2_EN,
speedPWM);//4
    digitalWrite(M3_IN1, LOW); digitalWrite(M3_IN2, HIGH); analogWrite(M3_EN,
speedPWM);//3
    digitalWrite(M4_IN1, LOW); digitalWrite(M4_IN2, HIGH); analogWrite(M4_EN,
speedPWM);//1
  }
}

void handleRoot() {
  String html = R"rawliteral(
<!DOCTYPE html>
<html>
<head>
  <meta charset="UTF-8">
  <title>ESP32 4WD Remote</title>
  <style>
    body { font-family: Arial; text-align: center; background: #f0f8ff;
padding: 20px; }
    .grid {
      display: grid;
      grid-template-columns: repeat(3, 80px);
      grid-template-rows: repeat(3, 80px);
```

```
      gap: 10px;
      justify-content: center;
      margin-bottom: 20px;
    }
    .btn {
      font-size: 32px;
      background: #4CAF50;
      color: white;
      border: none;
      border-radius: 10px;
      width: 80px;
      height: 80px;
      cursor: pointer;
      box-shadow: 2px 2px 6px rgba(0,0,0,0.3);
    }
    .btn:active { background: #2e7d32; }
    .stop { background: #f44336; }
    .stop:active { background: #b71c1c; }
  </style>
</head>
<body>
  <h2>ESP32 4WD Mecanum Remote</h2>
  <div class="grid">
    <button class="btn" onclick="send('forward_left')">↖</button>
    <button class="btn" onclick="send('forward')">↑</button>
    <button class="btn" onclick="send('forward_right')">↗</button>
    <button class="btn" onclick="send('left')">←</button>
    <button class="btn stop" onclick="send('stop')">■</button>
    <button class="btn" onclick="send('right')">→</button>
    <button class="btn" onclick="send('backward_left')">↙</button>
    <button class="btn" onclick="send('backward')">↓</button>
    <button class="btn" onclick="send('backward_right')">↘</button>
  </div>

  <h3>Rotation</h3>
  <button class="btn" onclick="send('rotate_left')">↺</button>
  <button class="btn" onclick="send('rotate_right')">↻</button>
  <h3>Speed Control</h3>
  <input type="range" min="0" max="255" value="150" id="slider">
  <p id="speedValue">Speed: 150</p>
  <script>
    function send(dir) {
      fetch("/move?dir=" + dir);
    }
    const slider = document.getElementById("slider");
    const speedValue = document.getElementById("speedValue");

    slider.oninput = function () {
```

```
      speedValue.innerText = "Speed: " + this.value;
      fetch("/speed?value=" + this.value);
    };
  </script>
</body>
</html>
)rawliteral";

  server.send(200, "text/html", html);
}

void handleMove() {
  String dir = server.arg("dir");
  move(dir);
  server.send(200, "text/plain", "OK");
}

void handleSpeed() {
  int val = server.arg("value").toInt();
  if (val >= 0 && val <= 255) speedPWM = val;
  server.send(200, "text/plain", "Speed set to " + String(speedPWM));
}

void setup() {
  Serial.begin(115200);

  pinMode(M1_IN1, OUTPUT); pinMode(M1_IN2, OUTPUT); pinMode(M1_EN, OUTPUT);
  pinMode(M2_IN1, OUTPUT); pinMode(M2_IN2, OUTPUT); pinMode(M2_EN, OUTPUT);
  pinMode(M3_IN1, OUTPUT); pinMode(M3_IN2, OUTPUT); pinMode(M3_EN, OUTPUT);
  pinMode(M4_IN1, OUTPUT); pinMode(M4_IN2, OUTPUT); pinMode(M4_EN, OUTPUT);
  stopAll();
  WiFi.begin(ssid, password);
  WiFi.setSleep(false);
  Serial.print("Connecting to WiFi");
  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }
  Serial.println("\nConnected! IP: " + WiFi.localIP().toString());
  server.on("/", handleRoot);
  server.on("/move", handleMove);
  server.on("/speed", handleSpeed);
  server.begin();
}
void loop() {
  server.handleClient();
}
```

Post Workshop Task (Club Activity):

1. Using the mecanum wheel, change the robot orientation to precise angle such as 15 degrees / 25 degrees etc and move the robot.
2. Find an interesting use case for the mecanum wheeled robot in your college context and share your build with the community of builders.